

Wolff Iteration Beyond Lattices: A Compatible Complemented-Dilation Theorem

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Status and source question

Status: candidate substantial partial result, likely valid. Egert and Kosmala [1] prove a Wolff-iteration theorem for A -convex quasi-Banach function spaces. On source PDF page 2, Remark 1.3 asks for the same conclusion for more general A -convex quasi-Banach spaces.

We then apply a reiteration result for Calderón products due to Gomez and Milman [9, Thm. 4.13], which provides the desired Wolff-iteration for quasi-Banach function spaces.

Theorem 1.2. *Let X_0, X_1, X_2, X_3 be A -convex quasi-Banach function spaces over a separable and σ -finite measure space Ω , where either X_0 or X_3 is separable and $X_0 \cap X_3$ is dense in X_1 and X_2 . Further, let $\theta, \eta, \lambda, \mu \in [0, 1]$ be parameters subject to the following conditions:*

- (i) $0 < \theta < \eta < 1$,
- (ii) $\theta = \lambda\eta$,
- (iii) $\eta = (1 - \mu)\theta + \mu$.

If $X_1 = [X_0, X_2]_\lambda$ and $X_2 = [X_1, X_3]_\mu$, then $X_1 = [X_0, X_3]_\theta$ and $X_2 = [X_0, X_3]_\eta$ hold up to equivalence of quasi-norms.

Remark 1.3. It remains an interesting open problem to prove Theorem 1.2 for more general A -convex quasi-Banach spaces.

We will recall the complex interpolation method for general quasi-Banach spaces

The theorem below gives the full Wolff conclusion for a class of spaces which need not themselves be function lattices: compatible common projection ranges of a function-space quadruple, provided the complementary quadruple obeys the same two local interpolation identities. The condition is automatic for a stationary complement.

Fix parameters

$$0 < \theta < \eta < 1, \quad \theta = \lambda\eta, \quad \eta = (1 - \mu)\theta + \mu. \quad (1)$$

All complex brackets below denote the outer complex method used in the source.

Proof intuition

A common bounded projection splits each ambient function space as $Y_j = X_j \oplus W_j$. Complex interpolation respects such finite direct sums. Thus, if both the desired range spaces X_j and their

complements W_j obey the two local identities, those identities recombine into the hypotheses of Egert–Kosmala’s function-space theorem for the Y_j . That theorem supplies the two global identities upstairs. Projecting back extracts exactly the desired identities for the possibly non-lattice spaces X_j .

The complement hypothesis is not bookkeeping: without information about the W_j , the two local identities for the X_j do not lift to the Y_j . This pinpoints what the projection argument can and cannot prove.

Complemented-dilation theorem

Theorem 1 (Compatible complemented-lattice dilation). *Let Y_0, Y_1, Y_2, Y_3 be A -convex quasi-Banach function spaces over one separable, σ -finite measure space. Suppose that either Y_0 or Y_3 is separable and that $Y_0 \cap Y_3$ is dense in Y_1 and Y_2 .*

Let P be a linear projection on a common ambient topological vector space containing the Y_j , and suppose that $P|_{Y_j}$ is bounded for $j = 0, 1, 2, 3$. Put

$$X_j = P(Y_j), \quad W_j = (I - P)(Y_j), \quad j = 0, 1, 2, 3,$$

with the inherited quotient (equivalently, subspace) quasi-norms. Assume, compatibly in the common ambient spaces, that

$$X_1 = [X_0, X_2]_\lambda, \quad X_2 = [X_1, X_3]_\mu, \quad (2)$$

$$W_1 = [W_0, W_2]_\lambda, \quad W_2 = [W_1, W_3]_\mu. \quad (3)$$

Then

$$X_1 = [X_0, X_3]_\theta, \quad X_2 = [X_0, X_3]_\eta \quad (4)$$

up to equivalence of quasi-norms.

Proof. Since P and $I - P$ are bounded on Y_j , the map

$$Y_j \longrightarrow X_j \oplus W_j, \quad y \longmapsto (Py, (I - P)y)$$

is an isomorphism with inverse $(x, w) \mapsto x + w$. These maps are compatible for all four indices, so we may identify

$$Y_j = X_j \oplus W_j \quad (5)$$

topologically.

We first record the elementary direct-sum identity

$$[A_0 \oplus B_0, A_1 \oplus B_1]_s = [A_0, A_1]_s \oplus [B_0, B_1]_s, \quad 0 < s < 1. \quad (6)$$

Indeed, an admissible function F for the couple on the left splits coordinatewise as $F = (F_A, F_B)$, where F_A and F_B are admissible for the two component couples. Conversely, two componentwise admissible functions combine into one. The boundary and ambient-sum quasi-norms are equivalent to the maximum (or any fixed finite direct-sum quasi-norm) of their component quasi-norms. Taking infima over functions with prescribed value proves (6) with equivalence of quasi-norms. This is also the usual retract principle for the complex method; compare Kalton–Mayboroda–Mitreia [2, Lemma 7.11].

Using (5), (6), and the two identities at parameter λ , we obtain

$$[Y_0, Y_2]_\lambda = [X_0, X_2]_\lambda \oplus [W_0, W_2]_\lambda = X_1 \oplus W_1 = Y_1.$$

The identities at parameter μ similarly give

$$[Y_1, Y_3]_\mu = Y_2.$$

All hypotheses of Egert–Kosmala’s Theorem 1.2 now hold for the function-space quadruple (Y_0, Y_1, Y_2, Y_3) . Hence

$$Y_1 = [Y_0, Y_3]_\theta, \quad Y_2 = [Y_0, Y_3]_\eta. \quad (7)$$

Finally, (6) applied to $Y_0 = X_0 \oplus W_0$ and $Y_3 = X_3 \oplus W_3$ shows, for every $0 < s < 1$, that

$$P[Y_0, Y_3]_s = [X_0, X_3]_s.$$

Applying P to the two identities in (7), and using $P(Y_1) = X_1$ and $P(Y_2) = X_2$, proves (4). $\square$

Corollary 2 (Stationary complement). *Under the hypotheses of Theorem 1, suppose that there is one quasi-Banach space W such that the four W_j are compatibly equal to W with equivalent quasi-norms. Then the local complement identities (3) are automatic, so the conclusion (4) follows from (2) alone.*

Proof. For every $0 < s < 1$, $[W, W]_s = W$ with equivalent quasi-norms. Constant admissible functions give $W \hookrightarrow [W, W]_s$, while the definition of admissibility and the maximum principle for the A -convex ambient space give the reverse inclusion. Substitution yields (3), and Theorem 1 applies. $\square$

Verification and scope

The proof is formal but not vacuous. A bounded projection range of a quasi-Banach function lattice need not be an order ideal and need not carry a compatible lattice structure, so the theorem reaches abstract A -convex spaces outside the literal hypotheses of the source theorem. The A -convexity of each X_j follows independently because it is isomorphic to a closed subspace of the A -convex space Y_j . Moreover, $P(Y_0 \cap Y_3) \subseteq X_0 \cap X_3$ is dense in X_1 and X_2 , so the density condition expected in the source question is inherited.

Each use of complex interpolation in the proof is justified directly from admissible functions. In particular, no vector-valued integration, duality, or unproved ordinary quasi-Banach reiteration theorem is being smuggled into the argument.

The result does not settle arbitrary A -convex quadruples. Eight focused upgrade attempts were made: gluing two local scales, applying the Kalton–Mitreá stability theorem, Banach-envelope/convexification reduction, constructing a non- A -convex-sum counterexample, passing to the inner method, common unconditional coordinates, complemented dilation, and finally removal of the complement hypothesis. The first six routes meet the same obstruction: for general quasi-Banach spaces the outer method need not be known to equal the core-generated inner method, and the Banach proof of reverse reiteration uses integration or duality unavailable here. The last route shows that a common projection alone supplies no identities among W_0, W_1, W_2, W_3 ; hence the complement condition cannot be deleted by this argument.

Bounded exact-title, arXiv-id, author, and keyword searches through 12 August 2026 used the phrases *Wolff-reiteration*, *A-convex quasi-Banach*, *inner complex interpolation*, *outer complex interpolation*, and *reiteration theorem*. They found no later general proof or counterexample. The theorem above appears to be a direct new corollary of known results rather than a new interpolation principle; novelty confidence is modest.

Human-review recommendation

Check that the four copies of P really are restrictions of one compatible projection in the intended application, rather than merely unrelated isomorphic projections. Then verify the two complement identities, or the stationary-complement condition. Those are the only application-specific points; the direct-sum and projection proof is otherwise exact. A full upgrade would require either a general reverse-reiteration theorem for the outer quasi-Banach complex method or a counterexample showing it fails.

References

- [1] M. Egert and B. W. Kosmala, *A Note on Complex Interpolation of Quasi-Banach Function Spaces*, arXiv:2412.12769 (2024); published version, *Studia Math.* (2026).
- [2] N. J. Kalton, S. Mayboroda, and M. Mitrea, *Interpolation of Hardy–Sobolev–Besov–Triebel–Lizorkin spaces and applications to problems in partial differential equations*, in *Interpolation Theory and Applications*, *Contemp. Math.* 445, Amer. Math. Soc., 2007, 121–177.
- [3] N. J. Kalton and M. Mitrea, *Stability results on interpolation scales of quasi-Banach spaces and applications*, *Trans. Amer. Math. Soc.* 350 (1998), 3903–3922.